Leadership & Consulting in Data Science

Week 3



Team Meeting – McDonald's EDA



Create the ***draft*** set of analyses to consider sharing with the client

- Consider importance of insights and relevance to problem
- Determine order…keeping storytelling in mind

Review the EDA generated by each of your team members

Create a list of questions that would help clarify your understanding further

Reconvene with the class and compare your key EDA findings with the class

- Create a short (1-3 visuals) from your EDA that tells a story
- A team member with the visuals should join the class Zoom session
- Each team will share their story (they do NOT have to be slides)
- Class should be prepared to provide feedback

Week 3 Agenda



Creating the Deck

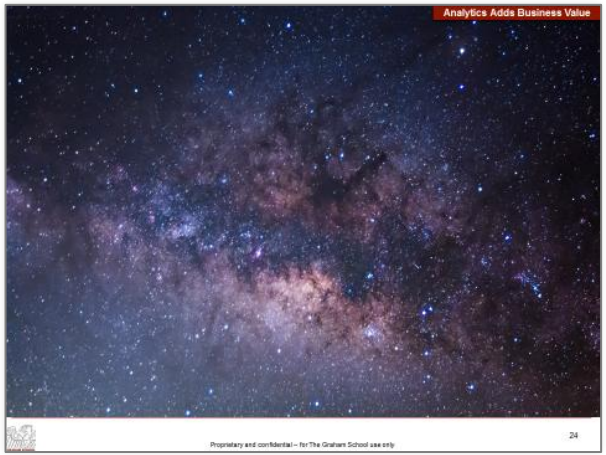
Presenting

Visualizations

Every presentation will be different…

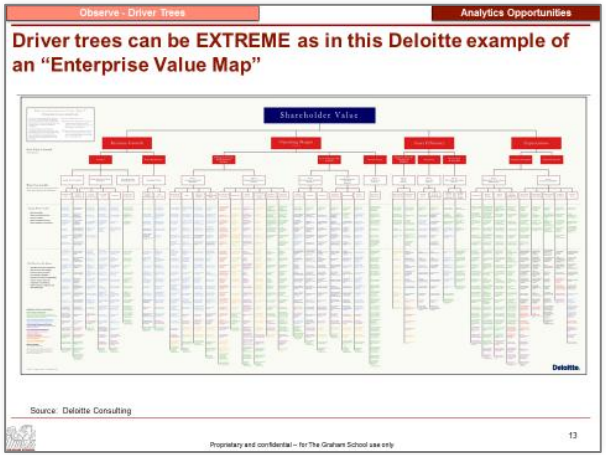
Strategically Communicating the Message

TEDx



Level of Detail

Extreme Detail



Will either extreme be appropriate for the CxO?

Building an effective presentation follows a logical process

PLAN

WRITE

1 What do I want to say?

Story

- Develop a bullet point outline of sections, pages, and detail
- Jump start this process with LLMs

2 How am I going to say it?

Storyboard

- Draw the high-level layout and images on each page
- Whiteboarding helps to develop custom messaging

3 How am I going to show it?

Slides

- Create the actual pages of the deck
- Work in PowerPoint in the final stage

Slide writing requires both content and a consistent structure and style

Content	Structure & Style
• Context • Story • Details	• Title Page • Agenda • Breadcrumbs • Page Numbers • References & Sources • Frameworks & Models • Examples, cases and illustrations • Visualizations • Introductions and Summaries • Appendices

For this class, for the senior executive, each slide MUST stand on its own!

Developing content requires addressing three (3) key dimensions

Dimension	Key Points	Relevance
 **Context**	Understanding the situation and tailor the document to the specific situation	Increases probability that the audience is receptive, and the message is well-received
 **Story**	Take key objectives and translate them into a logical flow; this is often called the **Storyline or Horizontal Logic**: Titles from each page can be compiled with the resulting text being the “story”	Focuses the output on a message that is rich, clear, relevant, and compelling
 **Details**	Breaking the storyline into simple and effective modules with details	Communicates complex messages logically, concisely and effectively

Tailor the Content to the audience and situation



Content Checklist

- [ ] **Desired outcome:** Buy in, understanding, urgency, faith
- [ ] **Seniority:** Core team members, management team, engineers, CIO
- [ ] **Power dynamics:** Alone, direct reports, peers, customers, bosses
- [ ] **Number:** 1, 3, 8, 120 attendees
- [ ] **Location:** Conference call, open office, hotel, board room, hallway
- [ ] **Occasion:** Introduction, proposal, kick-off, training, readout, final presentation
- [ ] **Style and culture:** Direct, analytical, hierarchical, risk averse, self-discovery
- [ ] **Language:** Taboos, terminology, politics, trusting, references
- [ ] **Duration:** 15 minutes, one hour, all day

In order to develop content, you must know what you want to say…you must have a story!



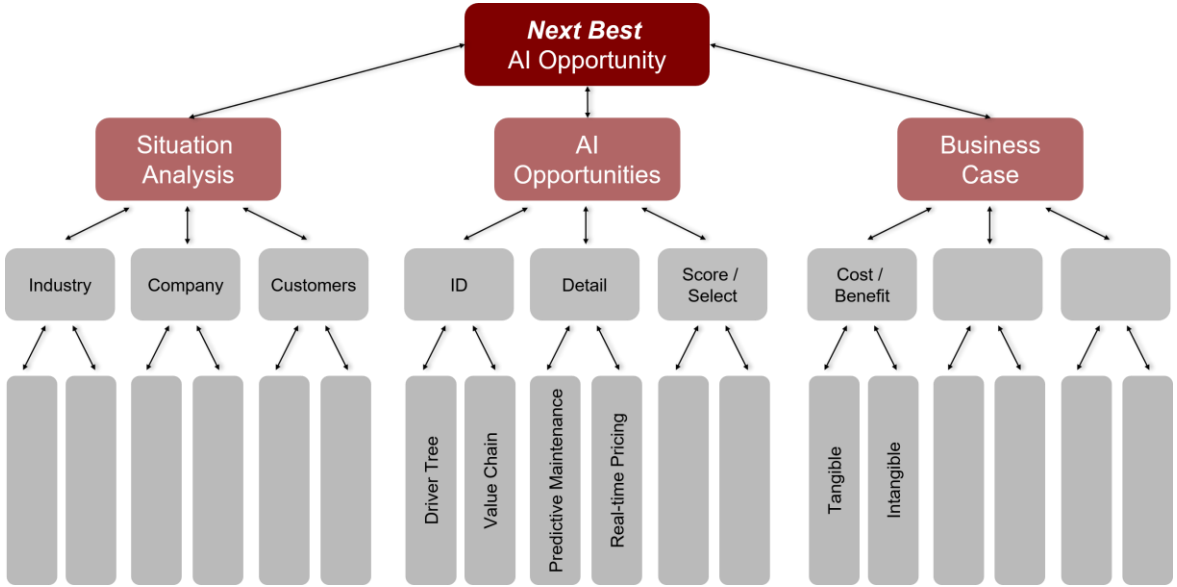
Story Guidelines

- Identify the objectives of the presentation
- State all the key messages to be delivered
- Categorize messages into groups
- Organize the groups and messages into a simple and logical flow

The written message should "pyramid" your details to focus on the key components of your story

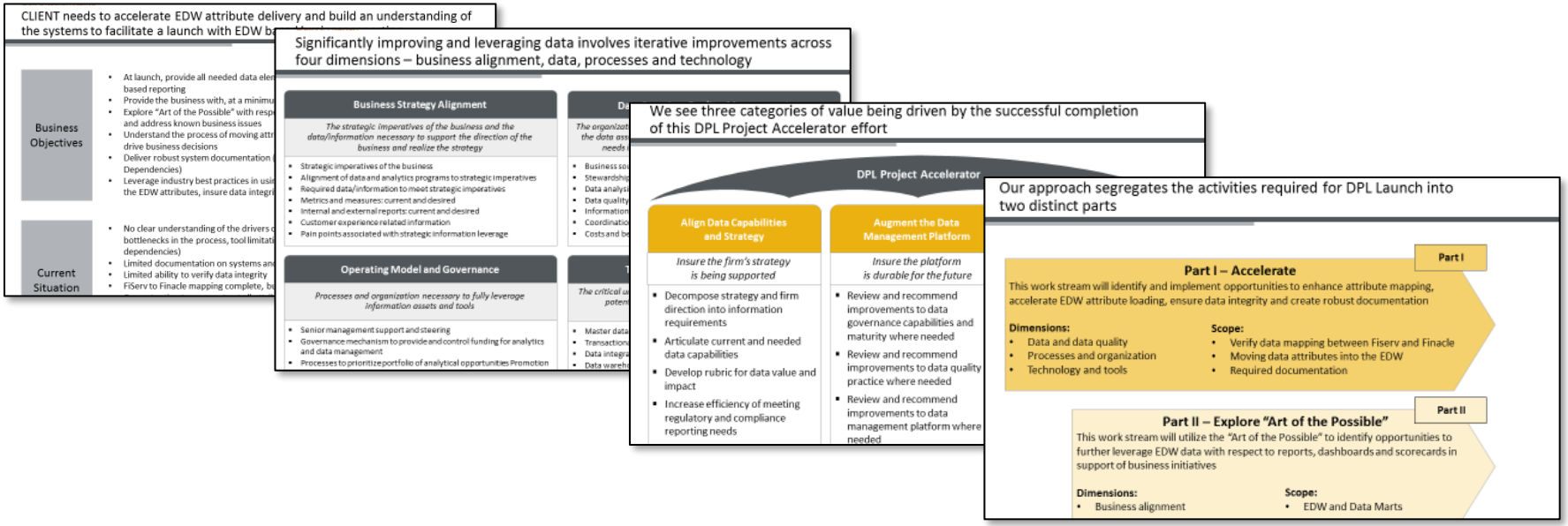
- Organize your thinking
  - Bottom up ↑
  - Top down ↓
- Group your ideas logically
- Test pyramid against basic rules

The Pyramid Principle drives storyline AND slide structure



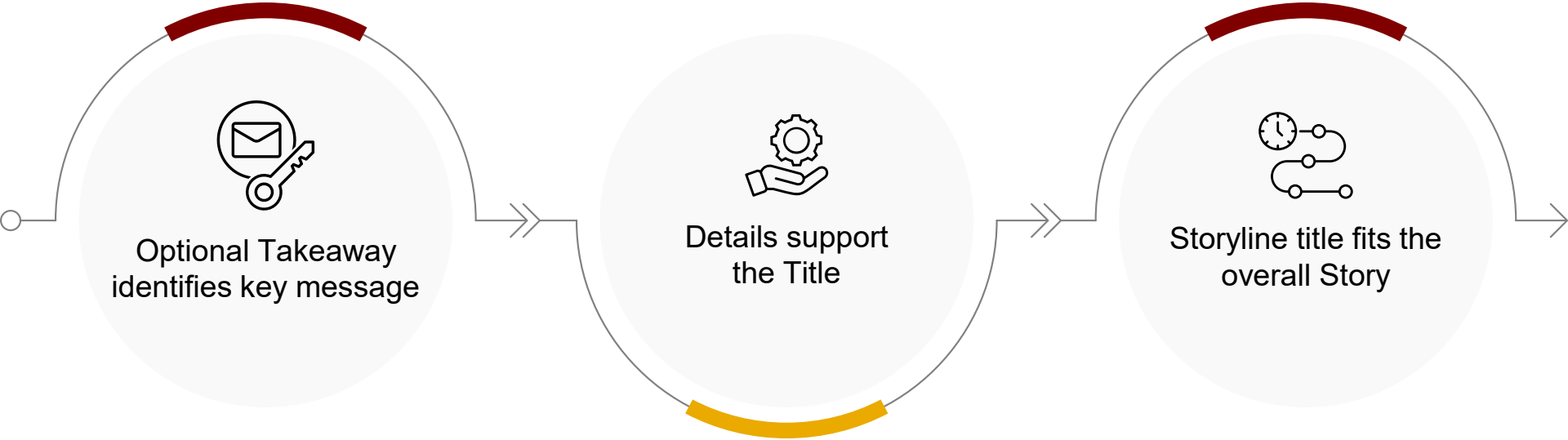
Source: Barbara Minto, The Minto Pyramid Principle, http://www.barbaraminto.com

For ***EVERY*** slide, a storyline title conveys part of the story



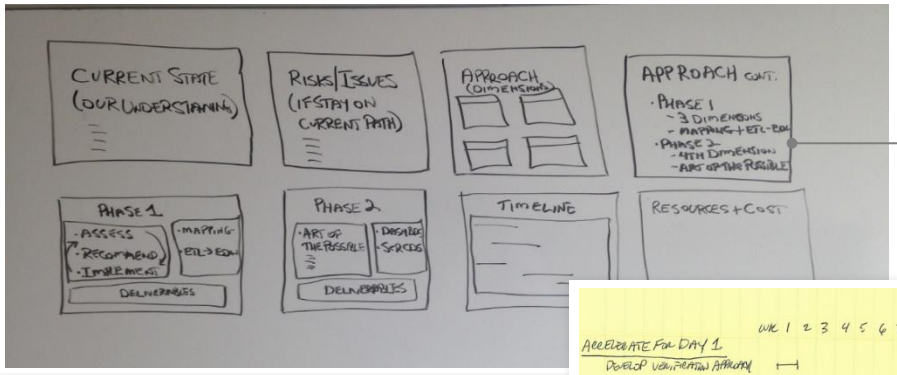
Together storyline titles, "Horizontal Logic," tell the story

For ***EVERY*** slide, the details, the optional takeaway and the storyline title support their part of the story

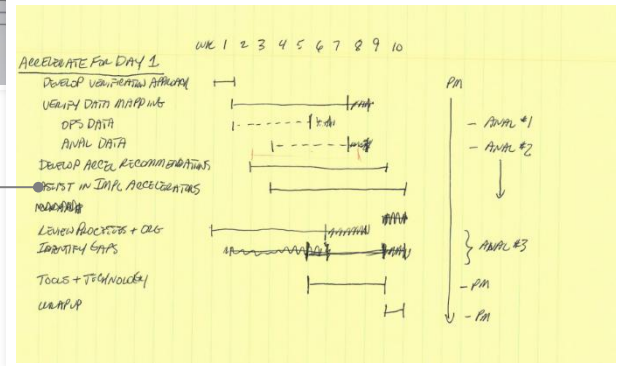


“Vertical Logic” provides each slide with a clear message

Example: A typical presentation starts with a whiteboard and other conversations…



Whiteboard captured on iPhone camera and shared with team



Scanned image of more detailed timeline developed during discussion

The Storyboard is driven by Pyramid thinking… driven by goals and clear understanding of the details required

The story translates key questions, objectives, answers, recommendations, and supporting analysis into a logical flow



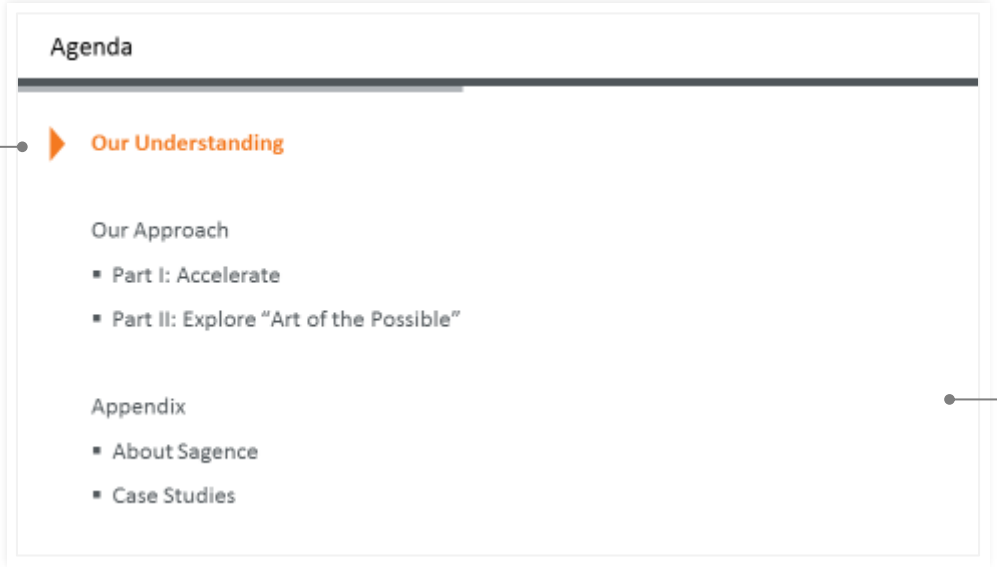
Story checklist

- ☐ **Clear, consistent theme:** One master theme supported across slides
- ☐ **Natural flow:** Story transitions across pages
- ☐ **Finite takeaways:** Specific messages designed to engage client(s)
- ☐ **Compelling:** Use the Pyramid Principle to structure the materials
- ☐ **Inductive reasoning:** Tell client up front what you are going to tell them
- ☐ **Horizontal logic / Storyline:** Logic consistent across slides
- ☐ **Vertical logic:** Logic consistent from header to supporting body text
- ☐ **Rich data:** Charts, graphs, tables, anecdotes, photos, quotes
- ☐ **Varied visuals:** Mixed visual layouts across slides (not repetitive)
- ☐ **Concise:** Presentation length appropriate to story

Think: “Enough, but no more!”

EVERY deck should have an agenda

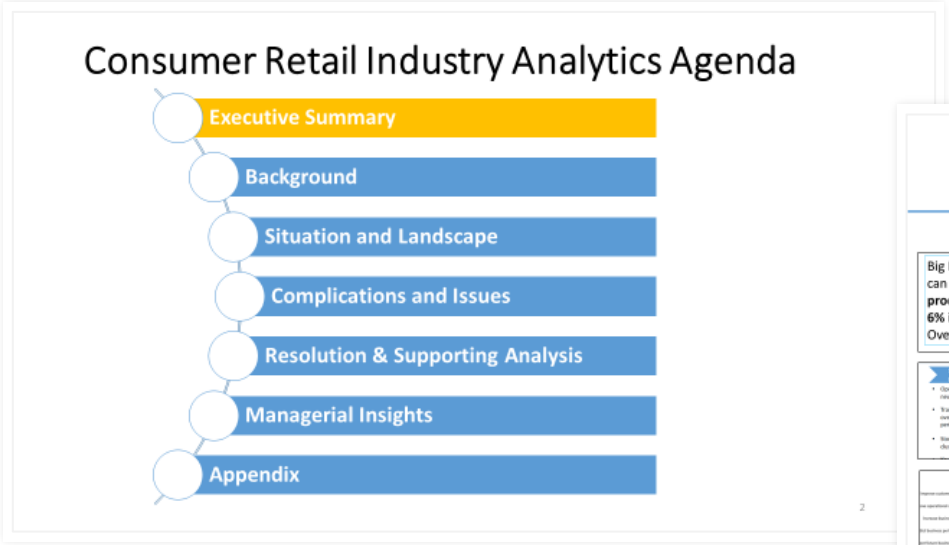
Give me an outline of the story – tell me what you're going to tell me!



Repeating this page tells me: where I've been, where I am and where I'm going

Unless your story line is crystal clear and relatively short, you should repeat the agenda and highlight where you are!

Use breadcrumbs to show your audience where you are in the story:





Page numbers are vital to reference location

Page Numbers provide the presenter and audience an easy reference to connect on location in the deck

References and sources on a page eliminate disruption of someone asking, "Where did this come from?"

Almost every slide should have sources

- Include: Document, Author, Date, Website
- Sources and References should be **consistently placed** and **consistently formatted**

Call out the date when relevant

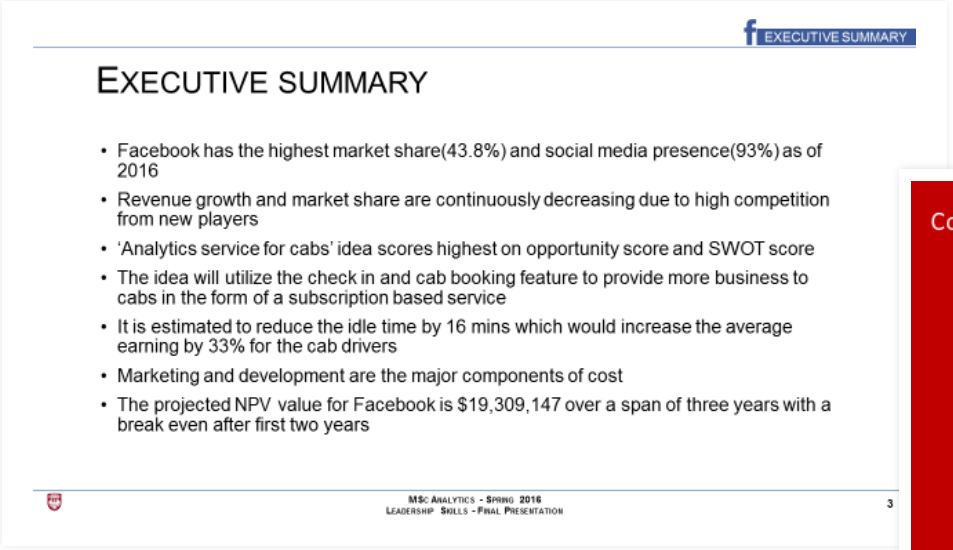
For websites, callout the name and give the URL

Call out "own analysis" to be clear where information was created

Source: A 2012 book; info source; Website name (http://...); Team Analysis

Footnotes: [1]McKinsey Book

Introductions, summaries or conclusions can be used in the beginning or end of the deck





Appendices provide a place for reference material



What to include…

- Could be more detailed information that wasn't deemed necessary for the main story / presentation
- Can be parallel information that was explored, but not considered necessary for the main story / presentation



Considerations…

- Slides should be structured within the Appendix
- Slides should have storyline titles
- In the real-world appendices typically are longer than the deck being presented

Maintain a checklist of what structure and styles you use in your decks



Structure & Style Checklist

- [ ] Title Page
- [ ] Agenda
- [ ] Page Numbers
- [ ] References & Sources
- [ ] Examples, cases and illustrations
- [ ] Visualizations
- [ ] Introductions and Summaries
- [ ] Appendices

Week 3 Agenda

Creating the Deck



Presenting

Visualizations

Three core speaking pillars to use every time you present

1. Tell the Story

- Tell them what you're going to tell them
- Tell them
- Tell them what you told them

2. Focus

- Relentlessly focus on the outcome

3. Know when to Stop

- Once you've reached your outcome, stop talking!



And remember…

Proper Planning and Preparation Prevents Poor Performance!

Read (research), Think, Speak

Be concise and precise

The Elements of Persuasive Argument



Focus on your goal **01**



Tailor argument to the decision-maker **02**



Build a Strong Argument **03**



Use delivery to enhance message **04**



Engage the Audience **05**

1. Focus on your goal



Primary Goal

- Clarify
- Be realistic
- Establish secondary and intermediate



Maintain Focus

- Avoid irrelevance
- Avoid tangents
- Allow spontaneity



Practice

- Expect questions and prepare answers
- Main point transitions
- Concede weak points

2. Tailor argument to the decision maker

- **Seniority:** Core team members, management team, engineers, CIO
- **Power dynamics:** Alone, direct reports, peers, customers, bosses
- **Number:** 1, 3, 8, 120 attendees
- **Location:** Conference call, open office, hotel, board room, hallway
- **Occasion:** Introduction, proposal, kick-off, training, readout, final presentation
- **Style and culture:** Direct, analytical, hierarchical, risk averse, self-discovery, formal
- **Language:** Taboos, terminology, politics, trusting, references
- **Duration:** 15 minutes, one hour, all day
- **Roles:** Facilitator, presenter, advocate, catalyst, challenger, attendee

People Strive to Remain Consistent with Previously Made Statements and Comments

Decider's Attitudes and Beliefs

01	02	03
Direction	**Intensity**	**Salience**
• Favorable	• Weak	• Relative importance
• Unfavorable	• Average	
• Neutral	• Strong	

Decider's Personality Type

Cognitive Thinkers / Right-Brain

Analytic	**Driver**
Amiable	**Emotive**

Affective Thinkers / Left-Brain

3.1 Build a strong argument by arranging the story

Introduction

Tell them what you are going to tell them

Make a Strong First Impression

- Make a startling statement
- Ask a question
- Use a quotation
- Tell a story
- State your purpose
- Pay a compliment
- Make a personal reference

Body

Tell them

Use Various Structures

- Chronology
- Evaluate alternatives
- Present problem and suggest solution
- Compare and contrast competing claims
- Keep points to 3 maximum
- Repeat key points

Conclusion

Tell them what you told them

Close with Passion

- Request the specific action you want
- Speak to heart of the argument

3.2 Build a strong argument with evidence, experience and strategy

Use Industry Examples



Relevant

So What

Compelling

Call to Action

Big idea

Thematic

3.3 Build a strong argument by cultivating ethos

Trust > Credibility

Elements of Ethos
Honesty
Integrity
Knowledge
Goodwill
Sincerity

People are persuaded by people they like

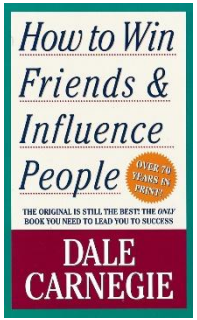
Ethos Tactics
Source Facts
Avoid Extreme Positions
Take Opposing Views Seriously
Organized Argument
Do Not Read Presentation
Smile, Be Courteous
Use Person's Name
Listen
Never talk down
Open with friendly conversation
Be yourself
Be dynamic
Be well-groomed

Watch & Listen



Think BIG EARS….
BIG EYES….
small mouth

“ The first rule of speaking is: Listen!
— Larry King



1. Smile
2. Show interest in other people
3. Encourage the other person to talk about himself
4. Make the other person feel important
5. Pay close attention to what the other person is saying

4. Use delivery to enhance message

Language and word choice

- Choose the appropriate language – avoid slang
- Word choice for your audience – avoid using technical language and acronyms

Clarity

- Speak to be understood
- Have a structure and flow
- Use examples

Volume

- Speak to be heard
- Project confidence

Pitch, tone, & pace

- Vary tone and pitch for impact
- Slow down – take your time to explain

Gestures

- Support message
- Avoid extraneous body movements

Facial expressions

- Smile
- Use when audience is talking to show interest

Body language and eye contact

- Maintain eye contact
- Good distance from audience
- Good posture – don’t cross arms

Transitions

- One continuous presentation
- Ride the coat tails of the person before you

Time

- Monitor time
- Know when to stop

5. Engage the Audience



People don't like to feel indebted

- Avoid demands and pleas
- Pace your argument to listener mood and mindset
- Allow participation
- Seek feedback
- Ask rhetorical questions
- Use anchors

Week 3 Agenda

Creating the Deck

Presenting



Visualizations

“There are three kinds of lies: lies, damned lies, and statistics”

-Benjamin Disraeli
British Prime Minister

“…but I’ll know it when I see it…”

-Potter Stewart
Supreme Court Justice

Take an audience perspective to improve your slides

Remember … they care about their world

Who are they?

How much detail does the audience need?

What do they care about in their role?

What are desired actions/decisions?

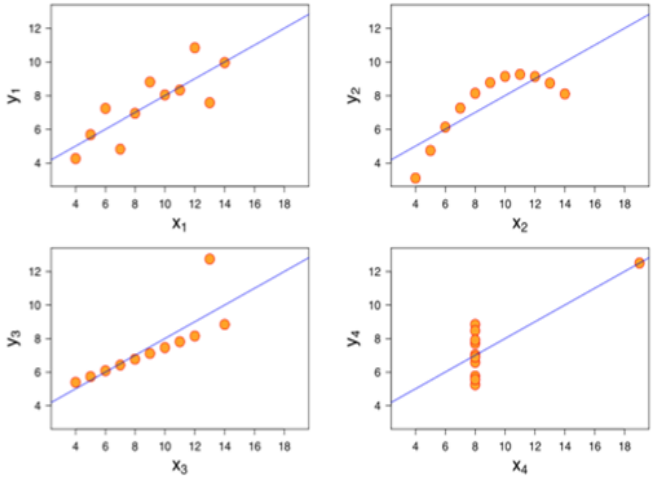
What visual would be easiest to understand?

Does the culture impact the visualizations?

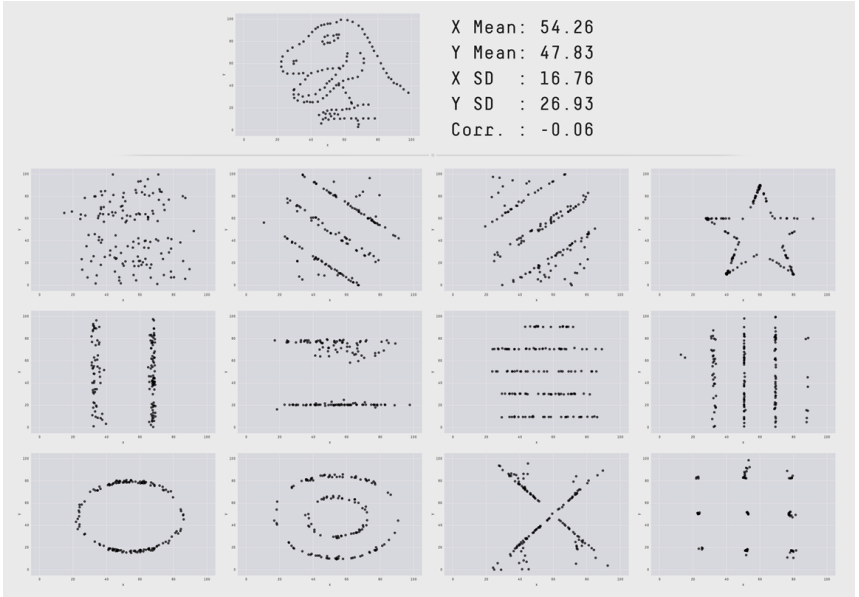
Numbers alone don't always convey the information correctly

Anscombe's Quartet

Property	Value
Mean of x in each case	9 (exact)
Variance of x in each case	11 (exact)
Mean of y in each case	7.50 (to 2 decimal places)
Variance of y in each case	4.122 or 4.127 (to 3 decimal places)
Correlation between x and y in each case	0.816 (to 3 decimal places)
Linear regression line in each case	$y = 3.00 + 0.500x$ (to 2 and 3 decimal places, respectively)

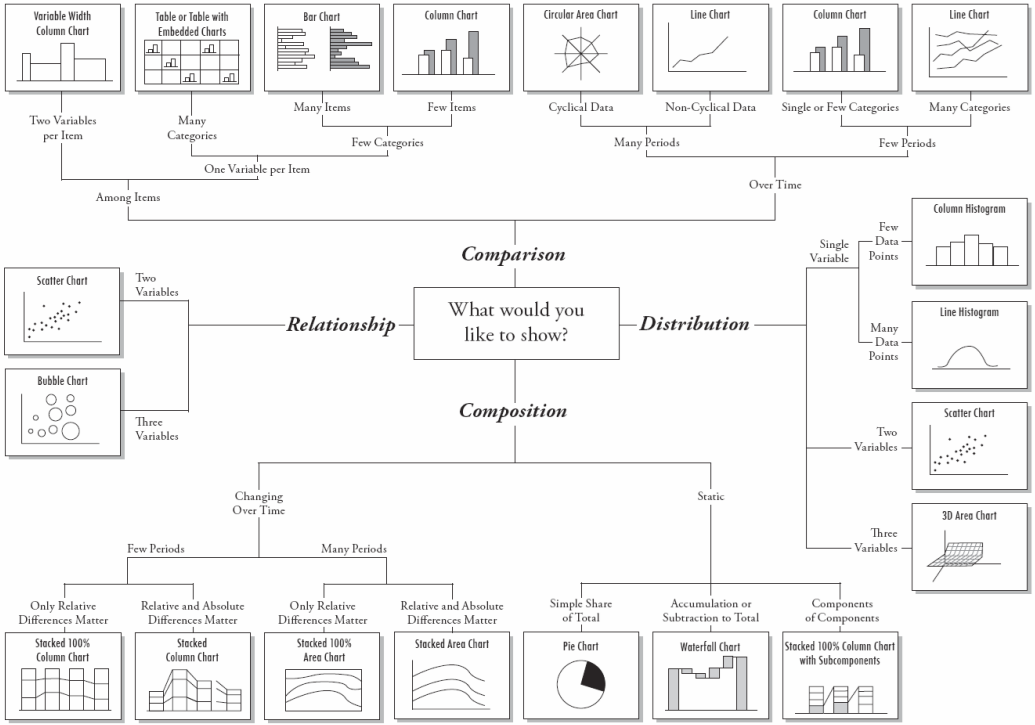


The Datasaurus Dozen



Source: https://www.autodeskresearch.com/publications/samestats

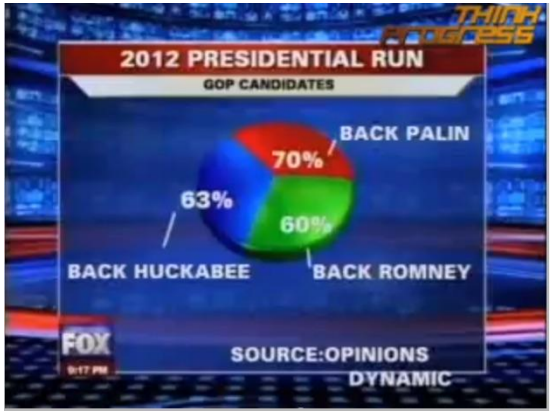
Choose the best chart type to convey the information



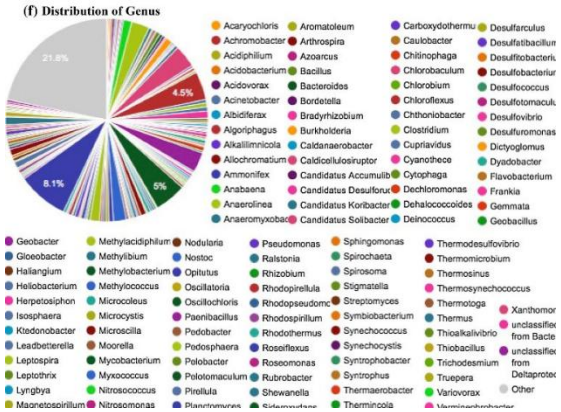
Source: http://www.labnol.org/software/find-right-chart-type-for-your-data/6523/

Make sure you convey the information correctly

Confusing!



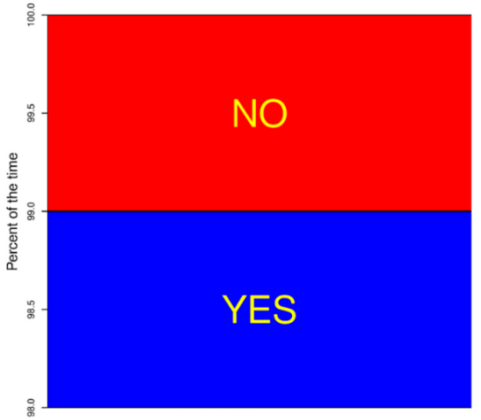
Unfathomable!



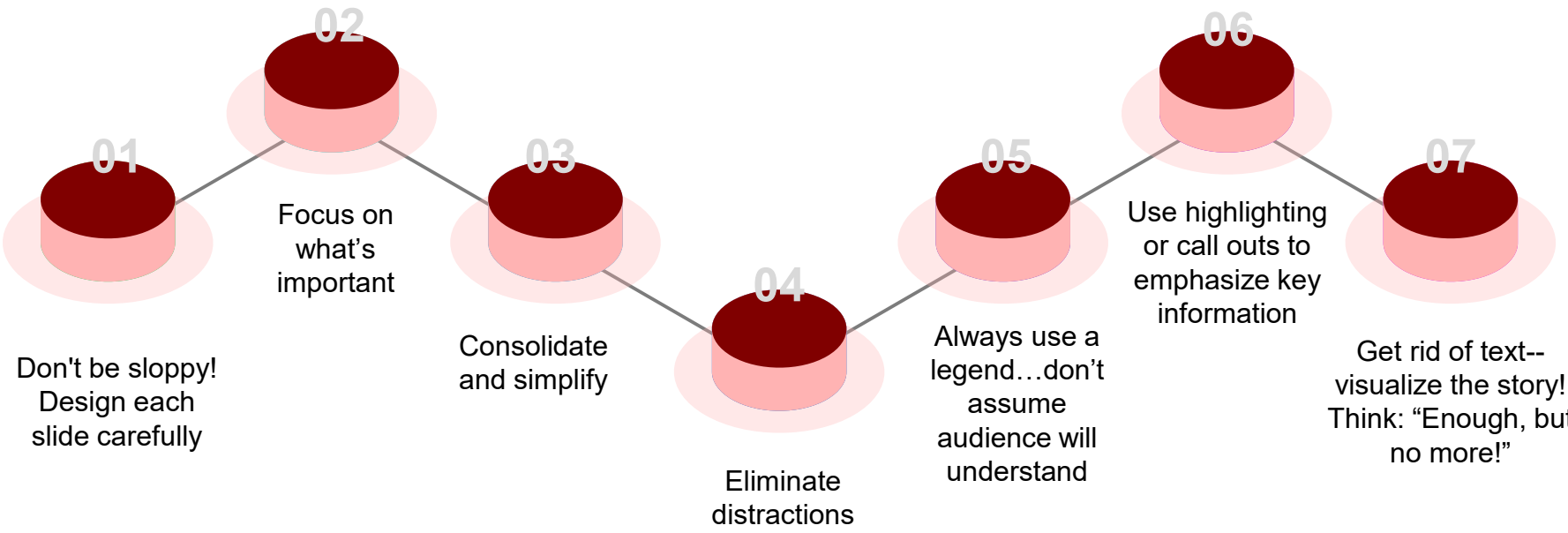
There are three kinds of lies: lies, damned lies and statistics

-Benjamin Disraeli
British Prime Minister

Misleading!



Develop “Best Practices” when communicating with slides



If you look at it and it doesn’t make immediate sense, then there’s something wrong!

Be on the lookout for typical visualization issues



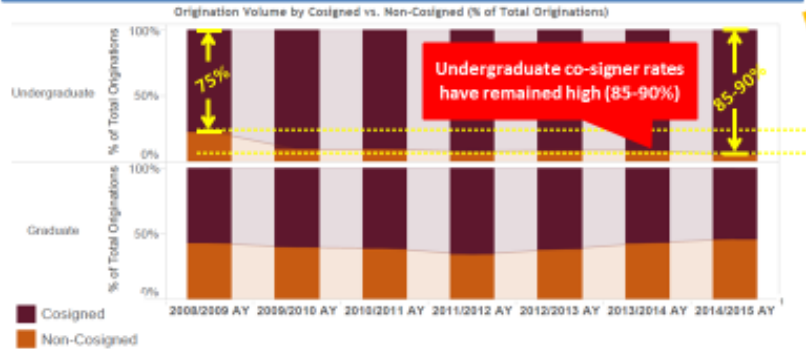
- Inappropriate display choice (e.g. pie charts)
- Variety for the sake of variety
- Too much information – visual clutter (e.g. 3d charts)
- Abuse of color (saturated, bright, inappropriate) – don't rely solely on color (color blind audience)
- Poor choices for item display
  - Noisy fill patterns
  - Noisy/heavy line styles
- Inconsistent ordering and placement
- Inconsistent or reversed scales
- Bad use of data
  - Encoding data inaccurately
  - Using counts vs. percentages
- Repeating words or patterns -- suggests reorganization

Visualization – example (student original)

OUR UNDERSTANDING: OPPORTUNITIES IN PRIVATE STUDENT LOAN MARKET

However 89% of overall private student loans required a co-signor. 94% of undergraduate and 56% of graduate loans

Cosigned vs. Non-cosigned Loan Distribution



Increase of cosigner rate within 6 years

10-15%

Address the secondary education funding gap through the increase of private student loan issuance. This can be accomplished **by targeting consumers who would otherwise be denied due to lack of co-signor or credit history**

- **Observation #1: Co-signor rates** consistently remained **high**, with rates rising from 75% for the AY2008/09 , remaining between 85% and 90% from that point onward
- **Observation #2: The primary driver** of the growing trend in co-signor rates is the strong increase in the percentage of cosigned loans for **undergraduate programs**, which remained above 90% since AY2009/10

Source: www.measureone.com

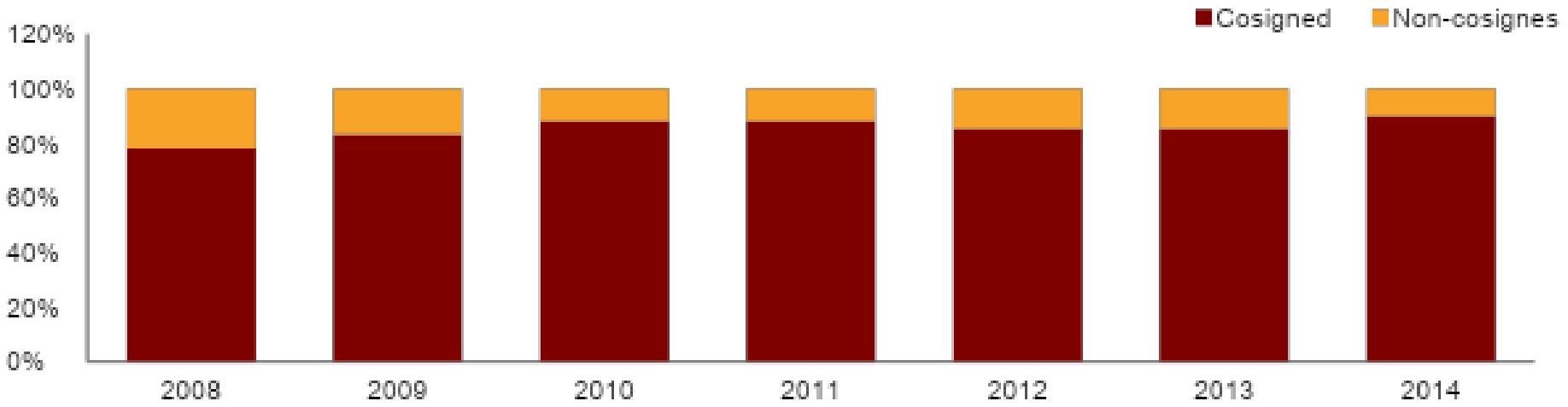


Visualization – example (student final)

Our Understanding: Background & Situation

Prevalence of cosigned funds for private student loans has increased from 75% in 2008 to 89.6% in 2014

Origination volume by Cosigned vs Non-cosigned



Private student loans can close the gap by targeting students who would otherwise be denied due to lack of co-signor or credit history

Source: XXX

Visualization – example (student original)

Problem | Solution | Approach | Costs | Implementation

Strategy and Implementation costs will be large for the first year and will taper off for maintaining the program

Implementation Plan

- 28 week engagement
- 6 weeks of strategy and planning
- 12 weeks of implementation
- 4 weeks of go-live/post go-live afterwork

Costs – 1st Year

- AWS cloud - $1.5m/year
- Data Brokers - $275,000
- Logistics - $75k
- Personnel - $750k



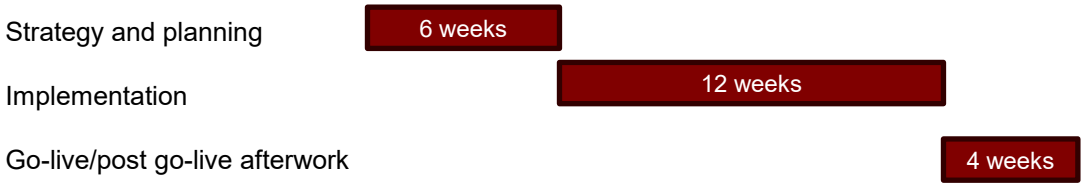
1 Article, Author, Publication, website, accessed 2/12/16

7

Visualization – example (student final)

Strategy and Implementation costs will be large for the first year and will taper off for maintaining the program

Implementation Plan (22 weeks)



Costs ($K)

	Year 1	Year 2	Year 3
AWS Cloud	$1,500	$1,500	$1,500
Data Brokers	$275		
Logistics	$75		
Personnel	$750		

Visualization – example (student original)

INDUSTRY BACKGROUND

Container shipping is a part of a larger value chain ripe with opportunities for analytics



1 Demand for foreign good. An order is placed, and the foreign manufacturer works with a freight forwarder to arrange transport from factory.

Analytics Application
Demand forecasting, carrier selection



2 Trucking company arrives at factory and loads order, along with many others, onto container.

Analytics Application
Space optimization



3 Freight forwarder determines optimal route for truck to take from factory to port. Shipping liner submits documentation about shipment to gov't authorities in both importing and exporting countries.

Analytics Application
Route optimization

4 

Container is loaded onto shipping liner, liner embarks for destination country.

Analytics Application
Space and route optimization, backhaul planning

5 

Before arrival, captain provides report to gov't of destination country containing info about ship, crew & cargo.

Analytics Application
Gov't data warehouse of value and tonnage of goods shipped

Source: World Shipping Council

INDUSTRY BACKGROUND

Container shipping as a part of a larger value chain ripe with opportunities for analytics



6 Ship receives clearance to arrive at port. Ship docks adjacent to cranes for unloading containers.

Analytics Application
None



7 Dockworkers, including crane operators, lashers, clerks, and cargo equipment operators, arrive to unload the ship.

Analytics Application
Staffing management



8 Customs officials may select specific containers for inspection.

Analytics Application
Staffing management



9 Workers load containers onto truck trailer or chassis for transport to destination.

Analytics Application
Space and route optimization



10 Truck arrives at import distribution center, where container is opened and contents are prepared for shipment.

Analytics Application
Order fulfillment

Source: World Shipping Council

Visualization – example (student final) – Option 1

Industrial background

Container shipping is a part of a larger value chain ripe with opportunities for analytics

01. Demand for foreign good. An order is placed, and the foreign manufacturer works with a freight forwarder to arrange transport from factory	**02.** Trucking company arrives at factory and loads order, along with many others, onto **container**	**03.** Freight forwarder determines optimal route for truck to take from factory to port. Shipping liner submits documentation about shipment to gov't authorities in both importing and exporting countries	**04.** **Container** is loaded onto shipping liner, liner embarks for destination country	**05.** Before arrival, captain provides report to gov't of destination country containing info about ship, crew & cargo	**06.** Ship receives clearance to arrive at port. Ship docks adjacent to cranes for unloading containers	**07.** Dockworkers, including crane operators, lashers, clerks, and cargo equipment operators, arrive to unload the ship	**08.** Customs officials may select specific containers for inspection	**09.** Workers load containers onto truck trailer or chassis for transport to destination	**10.** Truck arrives at import distribution center, where container is opened and contents are prepared for shipment
Analytics Application									
Demand forecasting, carrier selection	Space optimization	Route optimization	Space and route optimization, backhaul planning	Gov't data warehouse of value and tonnage of goods shipped	None	Staffing management	Staffing management	Space and route optimization	Order fulfillment

Source: womsaroone.com

Visualization – example (student final) Option 2

Industrial background

Container shipping is a part of a larger value chain ripe with opportunities for analytics

01. An order for a foreign good is placed, and the manufacturer works with a freight forwarder to arrange transport from factory

02. Trucking company arrives at factory and loads order, along with many others, onto **container**

03. Freight forwarder determines optimal route for truck to take from factory to port. Shipping liner submits documentation about shipment to gov't authorities in both importing and exporting countries

04. ★ **Container** is loaded onto shipping liner, liner embarks for destination country

05. ★ Before arrival, captain provides report to gov't of destination country containing info about ship, crew & cargo

06. ★ Ship receives clearance to arrive at port. Ship docks adjacent to cranes for unloading containers

07. Dockworkers, including crane operators, lashers, clerks, and cargo equipment operators, arrive to unload the ship

08. ★ Customs officials may select specific containers for inspection

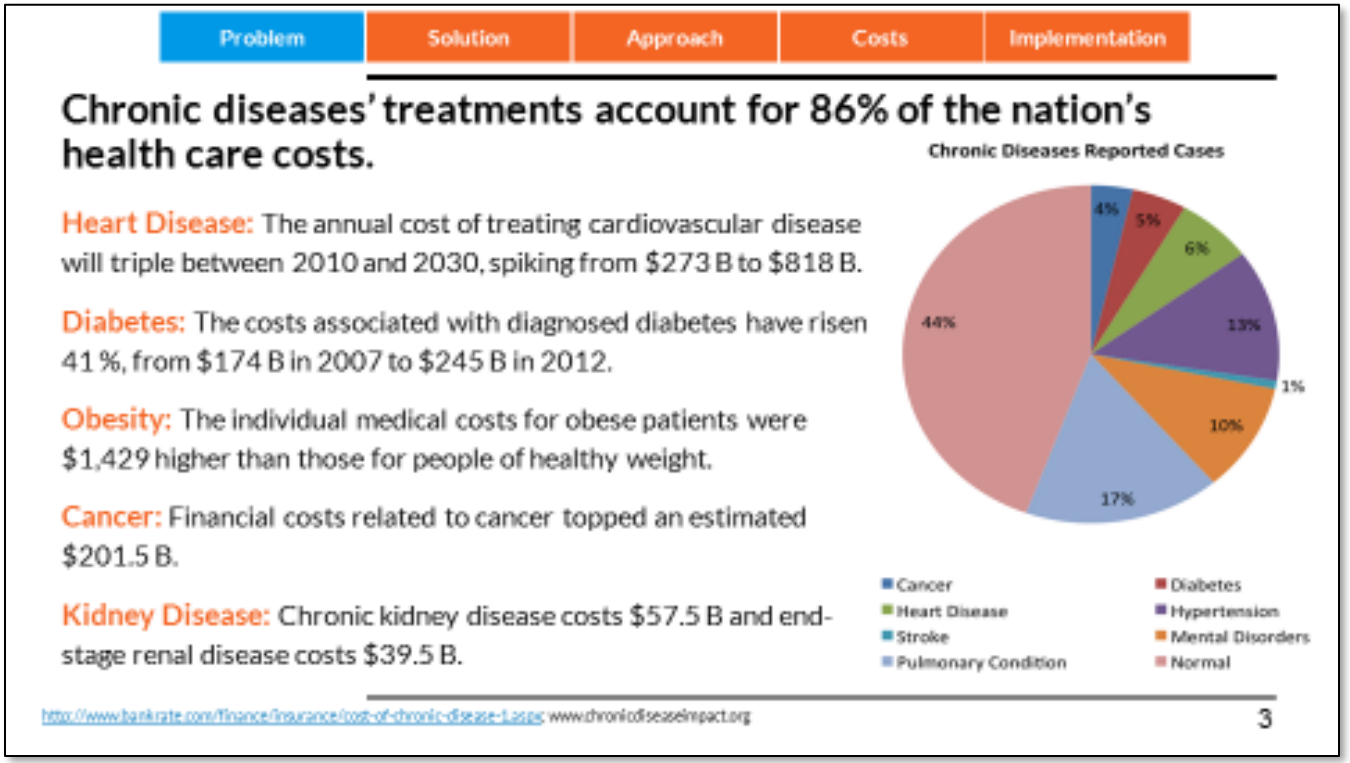
09. Workers load containers onto truck trailer or chassis for transport to destination

10. Truck arrives at import distribution center, where container is opened and contents are prepared for shipment

★ Container shipping direct involvement

Source: womsaroone.com

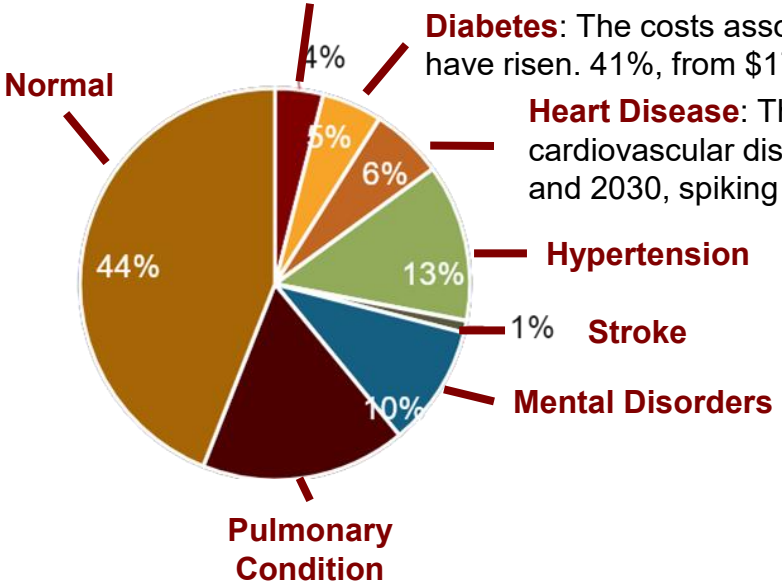
Visualization – example (student original)



Visualization – example (student final))

Chronic diseases' treatments account for 86% of the nation's health care costs.

Cancer: Financial costs related to cancer topped an estimated $201.5 B.



Diabetes: The costs associated with diagnosed diabetes have risen. 41%, from $174 B in 2007 to $245 B in 2012

Heart Disease: The annual cost of treating cardiovascular disease will triple between 2010 and 2030, spiking from $273 B to $818 B

Hypertension

Stroke

Mental Disorders

Other

Obesity: The individual medical costs for obese patients were $1,429 higher than those for people of healthy weight.

Kidney Disease: Chronic kidney disease costs $57.5 B and end-stage renal disease costs $39.5 B

Source: xxx

Visualization – example (student original)

28

Insurers have been a big part of this growth, the top 5 having $342 billion in total revenue and servicing over 150 million customers

- Domestic and international coverage
- Lives literally depend on efficiency and accuracy
- 25% of patients use 83% of resources



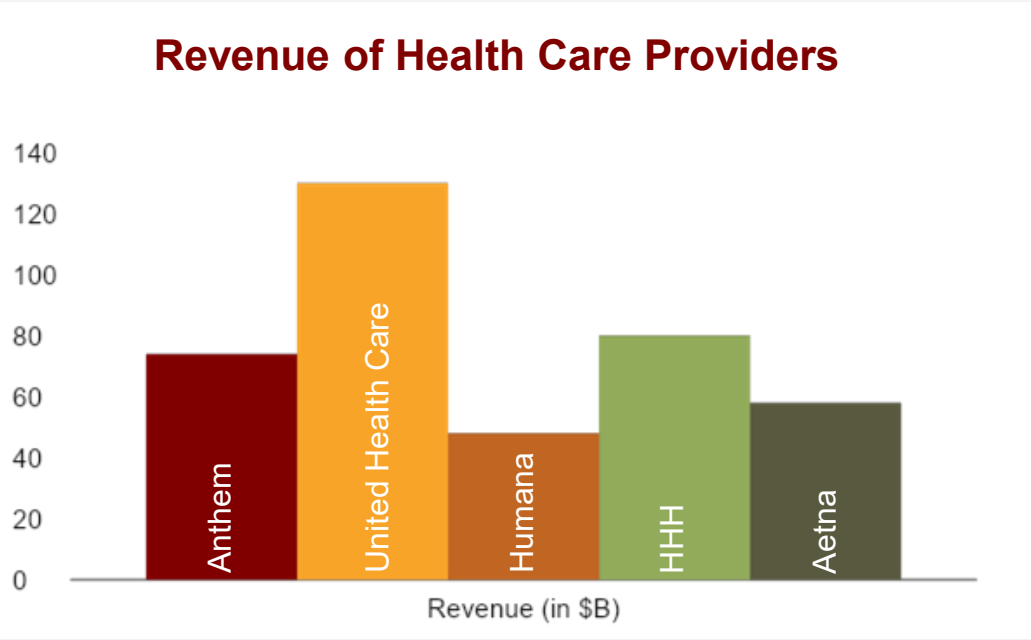
Visualization – example (student final)

Insurers have been a big part of this growth, the top 5 having $342 billion in total revenue and servicing over 150 million customers

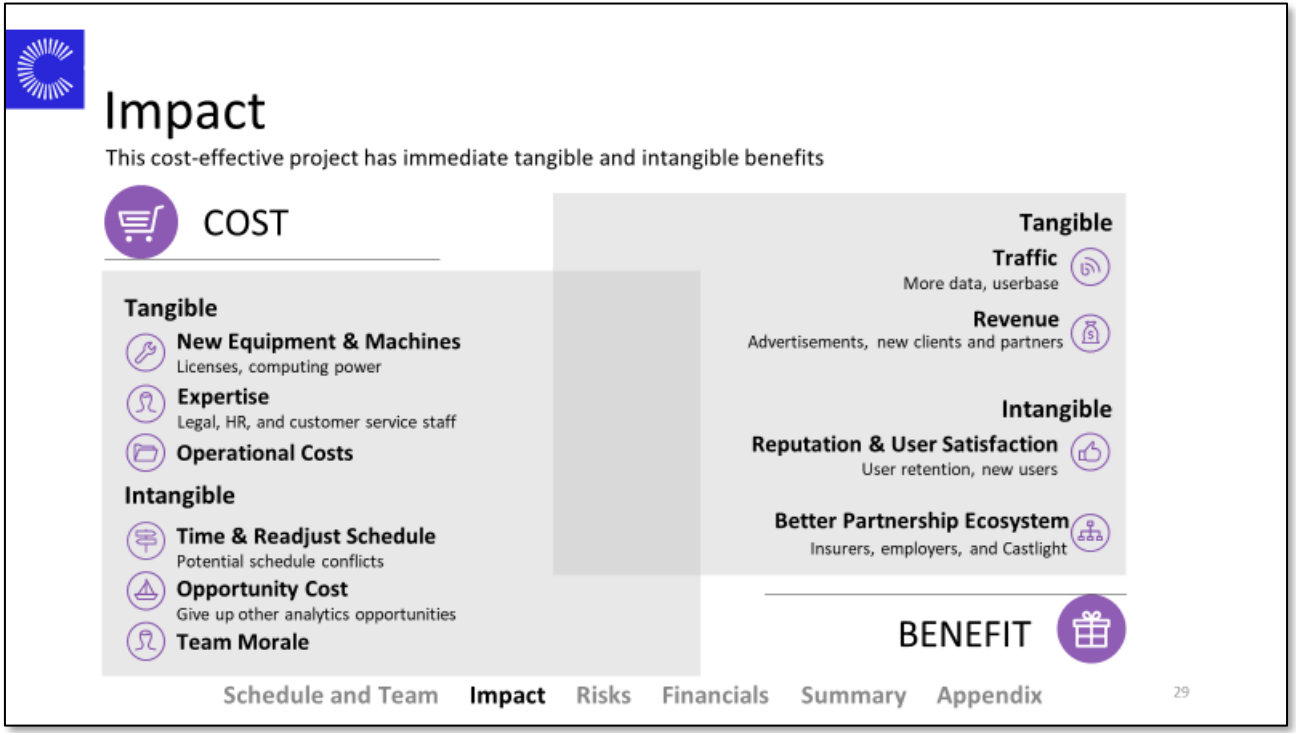
Domestic and international coverage

Lives literally depend on efficiency and accuracy

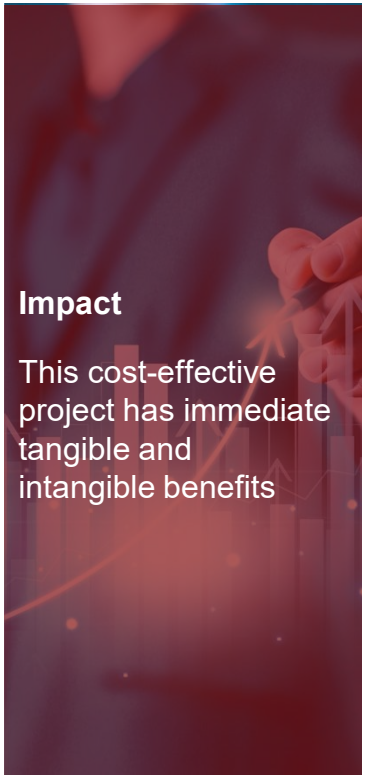
25% of patients use
83% of resources



Visualization – example (student original)



Visualization – example (student final)



	Cost	Benefits
Tangible	• **New Equipment & Machines** Licenses, computing power • **Expertise** Legal, HR, and customer service staff • **Operational Costs**	• **Traffic** More data, userbase • **Revenue** Advertisements, new clients and partners
Intangible	• **Time & Readjust Schedule** Potential schedule conflicts • **Opportunity Cost** Give up other analytics opportunities • **Team Morale**	• **Reputation & User Satisfaction** User retention, new users • **Better Partnership Ecosystem** Insurers, employers, and Castlight

Visualization – example (student original)



Visualization – example (student final)

We honed in to our best 3 opportunities and examined them

	Augmented scoring	Personalized Products	Voice Biometrics
Problem	• Outdated scoring model • Certain groups are locked out	• One-size-fits-all products being offered • Missed benefits	• PINs and other short, ineffective passwords • Outsiders can access info via Identity theft
Solution	• Expand model to include additional sources of data • Incorporate Marcus data	• Leverage customer data to offer personalized products, such as credit cards with special bonus	• Use a voice authentication model as an added layer of security
Benefit	• Better understanding of true customer risk • Tap into underserved market	• Increased customer retention, satisfaction • Increased market share in consumer retail	• Increased customer trust, confidence, ease of use, satisfaction • Increased security

Visualization – example (student original)

Financials

Limited up-front investment; revenue far exceeds the cost

Project Estimating and Approval	NPV						
		Amount	Year1	Year2	Year3	Year4	Year5
Project Investment							
	Hardware expenses (on-premise)	$100,000					
Project Expense							
	Concept Analysis	$100,000					
	Development	$1,500,000					
	Test & Validation	$200,000					
	Productionization	$70,000					
Total Project Investment		$1,970,000					
Operating Costs							
	Keep the lights on		$160,000	$160,000	$160,000	$160,000	$160,000
	Model Retraining and Validation		$80,000	$80,000	$80,000	$80,000	$80,000
	Product Marketing		$200,000	$200,000	$100,000	$80,000	$60,000
Total Operating Costs			$440,000	$440,000	$340,000	$320,000	$300,000
Project Benefits							
	Incremental Sales		$500,000	$1,000,000	$1,500,000	$2,000,000	$2,500,000
	Revenue from onboarding new insurers		$2,500,000	$2,500,000	$2,500,000	$2,500,000	$2,500,000
Total Benefits			$3,000,000	$3,500,000	$4,000,000	$4,500,000	$5,000,000
Net Yearly Cashflow			$2,560,000	$3,060,000	$3,660,000	$4,180,000	$4,700,000
Discount Rate	8%						

Net Present Value
$12,200,418.97

Schedule and Team | Impact | Risks | **Financials** | Summary | Appendix

27

Visualization – example (student final)

Financials Limited up-front investment; revenue far exceeds the cost

Project Estimating and Approval		Year 0	Year1	Year2	Year3	Year4	Year5
Project Investment	Hardware expenses (on-premise)	$100,000					
Project Expense	Concept Analysis	$100,000					
	Development	$1,500,000					
	Test & Validation	$200,000					
	Productionization	$70,000					
Total Project Investment		**$1,970,000**					
Operating Costs	Keep the lights on		$160,000	$160,000	$160,000	$160,000	$160,000
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	Revenue from onboarding new insurers		$2,500,000	$2,500,000	$2,500,000	$2,500,000	$2,500,000
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Net Yearly Cashflow			$2,560,000	$3,060,000	$3,660,000	$4,180,000	$4,700,000

Discount Rate: 8%

Net Present Value

$12,200,418.97

Team Meeting – McDonald's Presentation Prep



Prepare for your McDonald's presentation

Jumpstart the process with an LLM
- Assignment
- EDA Output
- Your story / storyline

Based on LLM and Team input create a rough draft of the deck

Create a workplan
- Who's working on what?
- Milestones
- When to meet
- Who's presenting what?
- Practice session

End of Deck